

Reduced-Order Dispatch Model for Simulating Marginal Emissions Factors for the United States Power Sector

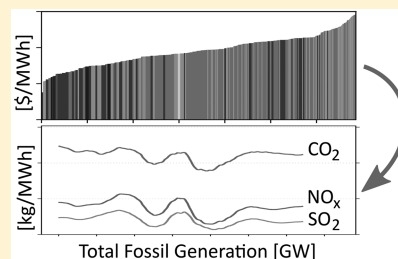
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Supporting Information

ABSTRACT: This study develops a reduced-order power plant dispatch model and uses it to simulate marginal emissions factors (MEFs) for the 2014–2017 United States (U.S.) electric grid at the North American Electric Reliability Corporation (NERC) regional level. MEFs help quantify the health, environmental, and climate change impacts caused by changes in marginal net electricity consumption, which could result, for example, from new technologies or policies. This study develops the model, validates it against historical data, and compares its simulated MEFs against historically derived regression-based MEFs. Our method accurately reproduces CO₂, SO₂, and NO_x emissions for multiple U.S. NERC regions and years and enables us to analyze future scenarios that are absent from the historical data. Though historically derived regression-based MEFs are generally more accurate, our simulated MEFs provide a more nuanced picture of how clusters of low- or high-emitting power plants of similar production cost create large swings in MEFs throughout the day. Policymakers could use these dynamic MEFs to target demand-reduction strategies at high-emissions portions of the power plant merit order.



INTRODUCTION

Growing interest in environmental sustainability has encouraged a variety of strategies aimed at reducing criteria air pollutants and greenhouse gas emissions in the power sector.^{1,2} Many strategies, such as renewable generation and electricity demand management, reduce emissions by displacing energy generation from fossil fuel power plants. Deciding which energy-displacing policies to prioritize requires an estimation of their emissions abatement potential. A rough estimation can be made using average emissions factors. Marginal emissions factors (MEFs), however, more accurately quantify emissions abatement potential by estimating a policy's impact on marginal generation—the set of power plants that would be most affected by an energy-displacing strategy.^{3–5}

Several methods have been developed to estimate the MEFs associated with electric grid operations.^{3,5–7} One such method performs regression analysis on historical data of power plant generation and emissions to show how changes in generation correlate with changes in emissions. With use of this method, MEFs can be readily calculated for numerous grid regions and for different levels of system demand, as illustrated by Siler-Evans' calculation of regional MEFs for the United States (U.S.).³ Once calculated, MEFs enable researchers to assess the short-term environmental impacts of renewable generation,^{4,8} electric vehicle charging,⁹ heating and cooling demand,¹⁰ energy storage,^{11,12} and other strategies for displacing conventional power generation.

Though straightforward and versatile, historical regression-based MEFs have limitations that narrow their application. First, they require high-resolution data on power sector operations

that may be unavailable. Second, their reliance on historical data limits their ability to represent long-term changes in the power sector, such as power plant retirements, that are absent from the historical data.^{7,11} While this regression-based approach is useful for assessing the effects of ongoing short-term strategies that influence marginal generation, it fails to accommodate changes in the generation fleet or to capture significant policy changes.⁵

Computational models, on the other hand, can analyze long-term scenarios by simulating the power plant dispatch.⁷ These types of models can explore the impacts of carbon reduction policy,^{13,14} large amounts of renewable generation,^{15,16} significant electric vehicle adoption,^{17,18} and other scenarios whose impact on the power plant dispatch is absent from the historical data.⁶ Though powerful, these dispatch tools can require substantial computational resources, specialized training, licensed software, and proprietary data. These requirements hinder their accessibility to general researchers and limit their flexibility for exploring broad power sector scenarios across multiple regions and time spans.

Since a user-friendly, freely available model for assessing future MEFs does not yet exist,⁷ researchers and policymakers seeking to quantify the environmental impacts of power sector policies must choose between historical regression MEFs that have limited application for long-term studies and complex dispatch models that might be difficult to use for broad scenario

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analysis. We provide an alternative method to address that gap—we simulate MEFs with an open-source, reduced-order dispatch model. Our model uses a merit-order-based, data-manipulation framework to quickly simulate the power plant dispatch and processes the output into MEFs. In contrast to existing methods, this framework enables us to generate a comprehensive set of simulated MEFs for the U.S. power sector covering multiple years and future scenarios for all eight electric grid regions in NERC (the North American Electric Reliability Corporation). Our results suggest that MEFs fluctuate more significantly than shown by regression-based methods. These fluctuating MEFs present opportunities to target demand-reduction strategies at high-emissions portions of the power plant merit order.

This article describes the methods and data used to develop our model, validates it with historical emissions data, and compares it to historical regression-based MEFs to reveal insights about power sector emissions intensity. In the [Results and Discussion](#) section, we first compare the model's 2017 emissions outputs with the historical data for all eight NERC regions. We then compute the marginal CO₂, SO₂, and NO_x MEFs for each region and compare them with historical data and regression-based MEFs from Azevedo et al.¹⁹ Last, we simulate all eight regions from 2014 to 2017 and quantify the annual error for each emissions type. We conclude by discussing the benefits and limitations of the regression-based¹⁹ and simulated methods.

METHODS AND DATA

This section describes the “Simple Dispatch” model, data, and calculations used to simulate and validate emissions for the U.S. power sector. Simple Dispatch is a merit-order-based model of the dispatchable-electricity-generating units in the United States. The model is open-source—it is built on publicly available data and its source code can be downloaded.²⁰ It uses input data to approximate the operating cost of each power plant in a regional grid. It dispatches those power plants in order of increasing operating cost to meet an exogenous, hourly, net demand time series for each NERC region (see [Supporting Information](#) for a map of the NERC regions' geographical boundaries). We use 2014–2017 as the reference years for our analysis. We exclude trade between NERC regions. The model outputs an hourly power plant dispatch that we use to estimate the total emissions and marginal emissions factors (MEFs) at different levels of net demand.

Merit Order Heuristic Model. Compared to complex, proprietary, optimization-based dispatch models, Simple Dispatch uses data manipulation and a reduced-order modeling framework to reduce solve time and complexity. It uses input data to calculate the merit order for each week of the year and uses those merit orders to dispatch the least-cost set of power plants for meeting the system's net demand during each hour of the year. The model can be used for broad scenario analyses, which include all types of dispatchable power plants. However, since the historical data we use for validation contains fossil fuel power plants only, the model's validation scenario in this study excludes nuclear, hydroelectric, geothermal, and biomass facilities. In this validation scenario, “net demand” equals total fossil fuel power plant generation and the model follows the procedure described here:

1. Read in data from

- EPA CEMS data set²¹—contains hourly electricity generation, emissions, and fuel consumption data for U.S. fossil fuel generators larger than 25 MW.
 - EPA eGrid data set²²—contains generator location, fuel type, capacity, and annual electricity generation, emissions, and fuel consumption for all U.S. generators.
 - EIA Form 923 data set²³—contains monthly fuel purchases and prices for most coal, natural gas, and oil fueled generators in the U.S.
2. We use EPA CEMS data²¹ to calculate the historical total hourly generation of the fossil fuel power plant fleet (i.e., the “net demand”).
 3. We preprocess the net demand profile to identify time periods when the minimum downtime constraint is active (see [Supporting Information](#), section IV).
 4. We use input data to calculate the weekly operating characteristics of each fossil fuel generator as follows:
 - (a) First, we calculate hourly heat rate [mmBtu fuel/MWh electricity] and hourly CO₂, SO₂, and NO_x emissions rates [kg/MWh] for each generator.
 - (b) Then we define each generator's weekly heat rate and emissions rates as the medians of its hourly heat rate and emissions rates.
 - (c) Finally, we define each generator's weekly capacity as its maximum weekly generation output. This captures events when generators shut down—for example, for maintenance—and their weekly generation output equals zero.
 5. We calculate each generator's production cost [\$/MWh] as heat rate × fuel price + variable operations and maintenance (VOM) cost.
 6. We arrange the generators in order of increasing production cost (i.e., the “merit order”).
 7. We solve the dispatch as follows:
 - (a) For each hour, we turn on all power plants in the merit order below the hourly net demand. The marginal generator—the most expensive online power plant—operates at part load.
 - (b) For each hour subject to the minimum downtime constraint, we resolve the dispatch using the minimum downtime constraint heuristic (see [Supporting Information](#), section IV).
 8. Finally, we return output data on total and marginal hourly generation cost, emissions, and fuel mix.

The above procedure describes the steps used to simulate the validation study presented in this article, which focuses on the operation of the fossil fuel generation fleet. If using the model for broader scenario analysis, we suggest the following adjustments:

1. The power plant fleet includes all dispatchable generators—not just fossil fuel power plants. Data on dispatchable, nonfossil fuel generators—including nuclear, hydro, geothermal, and biomass—come from EPA eGRID,²² which provides annual values for capacity, heat rate, and emissions rates.
2. For weeks when a generator's output equals zero (e.g., because of historical maintenance), assume its capacity equals the previous week's capacity. Though this adjustment ignores historical maintenance, it allows the model to turn on generators that were historically available for dispatch but were turned off because of their high production cost.

3. The net demand equals the total system demand minus the contribution of utility-scale wind and solar.

Fuel Price Data. Fuel price—a significant component of production cost—varies for each fuel, month, region, and generator. Regulated power plants report their monthly fuel prices in EIA Form-923.²⁴ This form contains monthly data on U.S. power plants' fuel purchases including volume, quality, and price. Since a generator might purchase multiple volumes of the same fuel at different prices, we calculate each generator's monthly fuel price as the sum of its different prices weighted by purchasing volume. For most regions, EIA Form-923 provides fuel price data for the majority of generators listed in CEMS. For generators without EIA Form-923 fuel price data, we assign the monthly fuel price profiles of other generators of the same type in the same NERC region, uniformly distributing the profiles starting with the largest volume purchasers. These assignments assume that power plants within the same NERC region will have similar fuel price profiles to each other.

We subdivide natural gas purchases according to purchase type—spot prices, contracts, and tolling agreements. A generator without fuel price information that purchases its natural gas via spot prices, for example, will be assigned the monthly fuel profile of a power plant in the same region that purchases natural gas via spot prices. EIA Form-923 excludes fuel prices for tolling-agreement natural gas purchases. We assume tolling-agreement fuel prices will be similar to contract-type fuel prices.

The U.S. power sector has relatively few lignite coal plants, and many regions have insufficient lignite fuel price data to adequately populate the model. We assign the average national monthly lignite prices to any lignite plants without fuel price data.

For fuels without EIA Form-923 data, we use commodity prices reported by the Energy Information Administration.²⁵

Variable Operation and Maintenance. Variable operation and maintenance costs (VOM) describe the generators' variable, nonfuel expenses for producing power. These VOM costs vary by power plant depending on prime mover, age, and fuel type.²⁶ We use data from "The western wind and solar integration study phase 2" published by the National Renewable Energy Laboratory,²⁶ which provides a range of VOM costs for different generator types. We assume the variation in VOM values can be explained by the variation in power plant age. We assign VOM costs of 3.91 and 4.52 \$/MWh, for example, to 40- and 50-year old coal plants. Refer to the [Supporting Information](#) for more detail on how we calculate VOM for each power plant type.

Minimum Downtime Constraint for Coal. Since thermal power plants and emissions technologies must follow warm-up and cool-down procedures when cycling on and off, it can be advantageous for a generator to sell power at a loss for short time periods to avoid the cost and lost time caused by a shut-down—start-up procedure.²⁷ These "minimum downtime" considerations can especially affect coal power plants operating at lower levels of net demand. The Simple Dispatch model approximates this effect using a minimum downtime constraint. Under this constraint, if a drop and rise in demand would cause a marginal coal plant to turn off and on within a 12-h period,²⁸ that coal plant will operate at its minimum output—40% of capacity²⁹—instead of turning off. Refer to the [Supporting Information](#) for more detail.

Combined-Heat-and-Power Plants. Many regions contain combined-heat-and-power (CHP) plants, which produce electricity as well as heat for supplying nearby thermal demands. These CHP facilities report their gross load—heat plus electricity—to the EPA CEMS database.³⁰ Since our model focuses on the power sector, we exclude each CHP plant's thermal production by reducing the CEMS-calculated capacity used in our model. We reduce CHP capacity by using the "generation and fuel data" from EIA Form-923.²⁴ For any generator flagged as a CHP plant, we divide its annual fuel consumed for electricity generation by its annual total fuel consumption. We multiply this ratio—which ranges from 0 to 1—by the CEMS-calculated capacity to reduce the CHP plant's modeled capacity to reflect its electricity generation activity.

Results Calculations. Running the Simple Dispatch model for an hourly resolution, 1-year, net demand time series yields an hourly power plant dispatch that can be used to calculate total and marginal emissions of CO₂, SO₂, and NO_x. [Equation 1](#), for example, equates hourly total CO₂ emissions to the sum of the electricity output of each generator g multiplied by its emissions rate.

$$\text{CO}_{2\text{total}}(t) = \sum_g \left([\text{MWh}](g, t) \times \left[\frac{\text{kg of CO}_2}{\text{MWh}} \right](g, t) \right) \quad \forall g \quad (1)$$

We calculate marginal emissions factors (MEFs)—the change in emissions caused by a change in demand—as the derivative of the curve for total emissions versus total fossil generation. First, we sort the hourly total emissions by increasing amounts of fossil fuel generation. Then we take a rolling average of that data to define the curve of total emissions versus total fossil generation. Second, we define the MEFs by estimating the derivative for each total fossil generation—that is, "net demand"—value of that curve. [Equation 2](#), for example, equates the CO₂ MEF for each net demand value d as the change in total CO₂ emissions divided by the change in d between 438 data points (5% of the data) above d to 438 data points below d . We take the rolling average of those hourly MEF calculations to define the curve of MEFs versus total fossil generation. See [Figure 2](#) in the [Results and Discussion](#) section for an example of the total emissions and MEF curves.

$$\text{MEF}_{\text{CO}_2}(d) = \frac{\text{CO}_{2\text{total}}(d + 438) - \text{CO}_{2\text{total}}(d - 438)}{(d + 438) - (d - 438)} \quad (2)$$

We compare our simulated results against the historical hourly power plant dispatch from the EPA CEMS data set.²¹ We quantify the relative error of the output results by dividing the absolute error by the sum of the historical values. [Equation 3](#) shows this calculation. When the total emissions error is calculated, $X_{\text{historical}}$ equals the hourly historical emissions and $X_{\text{simulated}}$ equals the rolling average of the simulated curve for total emissions versus total fossil generation. When the MEF error is calculated, $X_{\text{historical}}$ equals the rolling average of the derivative of the historical curve for total emissions versus total fossil generation and $X_{\text{simulated}}$ equals the rolling average of the derivative of the simulated curve. We also compare our results and error against historically derived, regression-based MEFs provided by Carnegie Mellon University's Center for Climate and Energy Decision Making.¹⁹

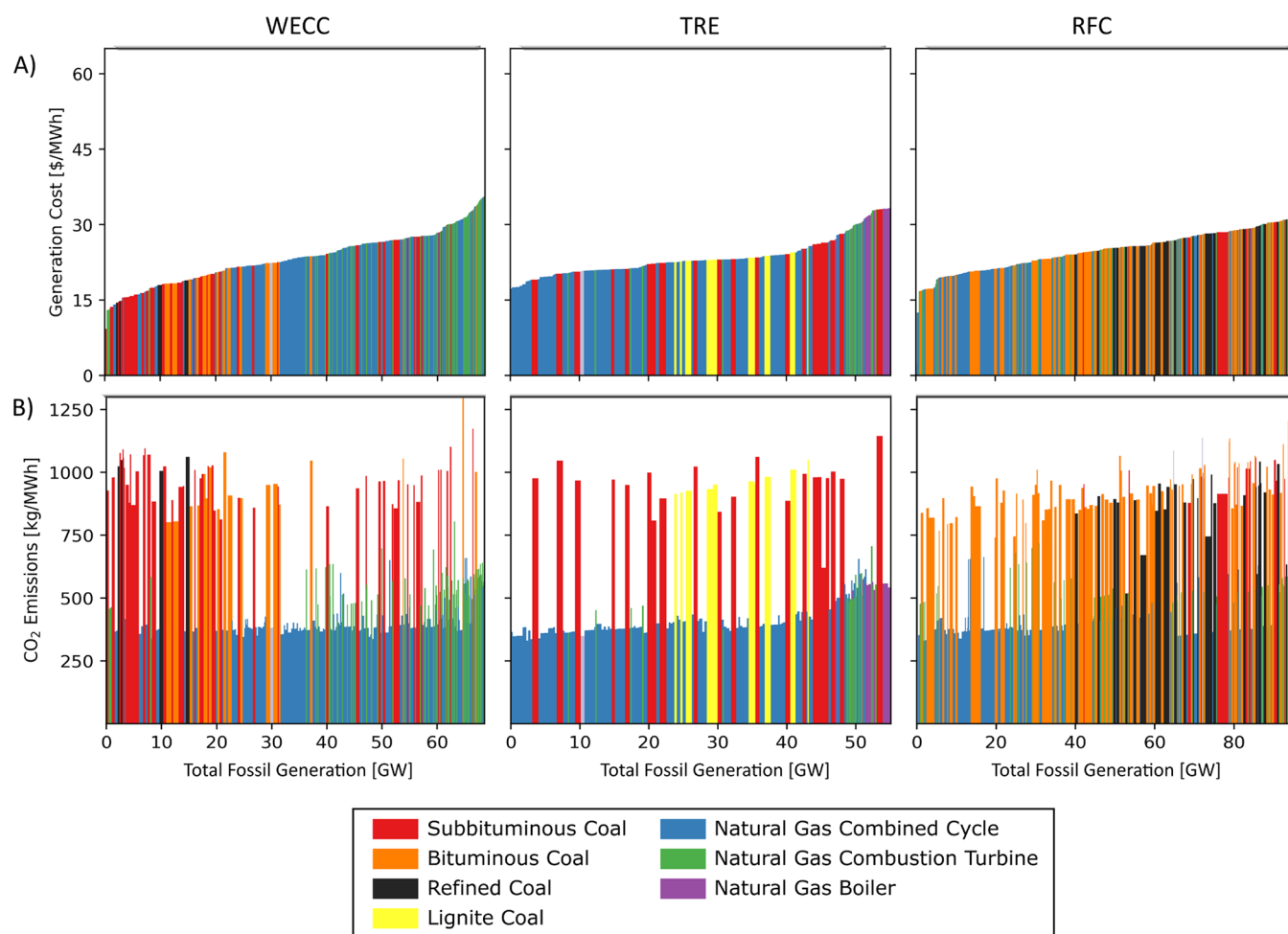


Figure 1. Merit order—ascending in order of operation cost—for the first week of August for three NERC regions showing (A) generation cost and (B) CO₂ emissions rates. Note that (A) and (B) have the same ordering of power plants.

$$\text{Error}_X = \frac{\sum_{t=0}^{8759} |X_{\text{historical}}(t) - X_{\text{simulated}}(t)|}{\sum_{t=0}^{8759} X_{\text{historical}}(t)} \quad (3)$$

Model Applications and Limitations. We recommend using our method for sensitivity analyses of broad trends affecting the economic dispatch of regional-scale power plant fleets. The method can analyze scenarios that affect the merit order—for example, power plant retirements, changing fuel prices, and carbon taxes. It can also be used for studying large-scale changes in electricity demand—for example, demand response, technology adoption, and energy efficiency. Analysis results should focus on broad trends in power sector emissions—for example, changes in annual emissions and trends in marginal emissions factors. We expect the accuracy of these types of analyses to be similar to the validation results presented in this study.

Researchers using the tool should keep in mind two main limitations. First, the Simple Dispatch tool omits transmission. At the subregional level, the model assumes unlimited transmission capacity—that is, any power plant in the regional fleet can supply any load within that region. Transmission congestion in real-world systems causes suboptimal power plant dispatch when expensive plants near the load are dispatched before inexpensive plants separated from the load by congested transmission lines. Consequently, the Simple Dispatch model will yield a lower-cost dispatch solution than the historical data.

At the inter-regional level, the model excludes trade between NERC regions. The assumption holds for this validation study, which reproduces the actual fossil generation for each region. For scenario analyses, however, a lack of inter-regional trade will yield a higher-cost dispatch solution than the historical data. Historical data (see [Supporting Information](#)) indicate that most NERC regions trade less than 5% of their annual demand. Exceptions include NPCC, which imports 15% of its annual demand from Canadian hydropower, and FRCC, which imports/exports 15% of its annual demand.

Second, the Simple Dispatch tool omits wind and solar curtailment. This omission might impact the accuracy of simulated MEFs. Research on the relationship between renewables curtailment and MEFs remains nascent; some studies suggest a negligible impact³¹ while others show increasing contributions from wind and solar to marginal generation in some regions.⁸ Though the Simple Dispatch model excludes the market structures, forecast errors, and transmission congestion that cause wind and solar to operate on the margin, its simulation output could be merged with estimations of the marginal contributions of renewable generators. The simulated MEFs from this study, for example, could be balanced with hourly marginal renewable data using a weighted average.

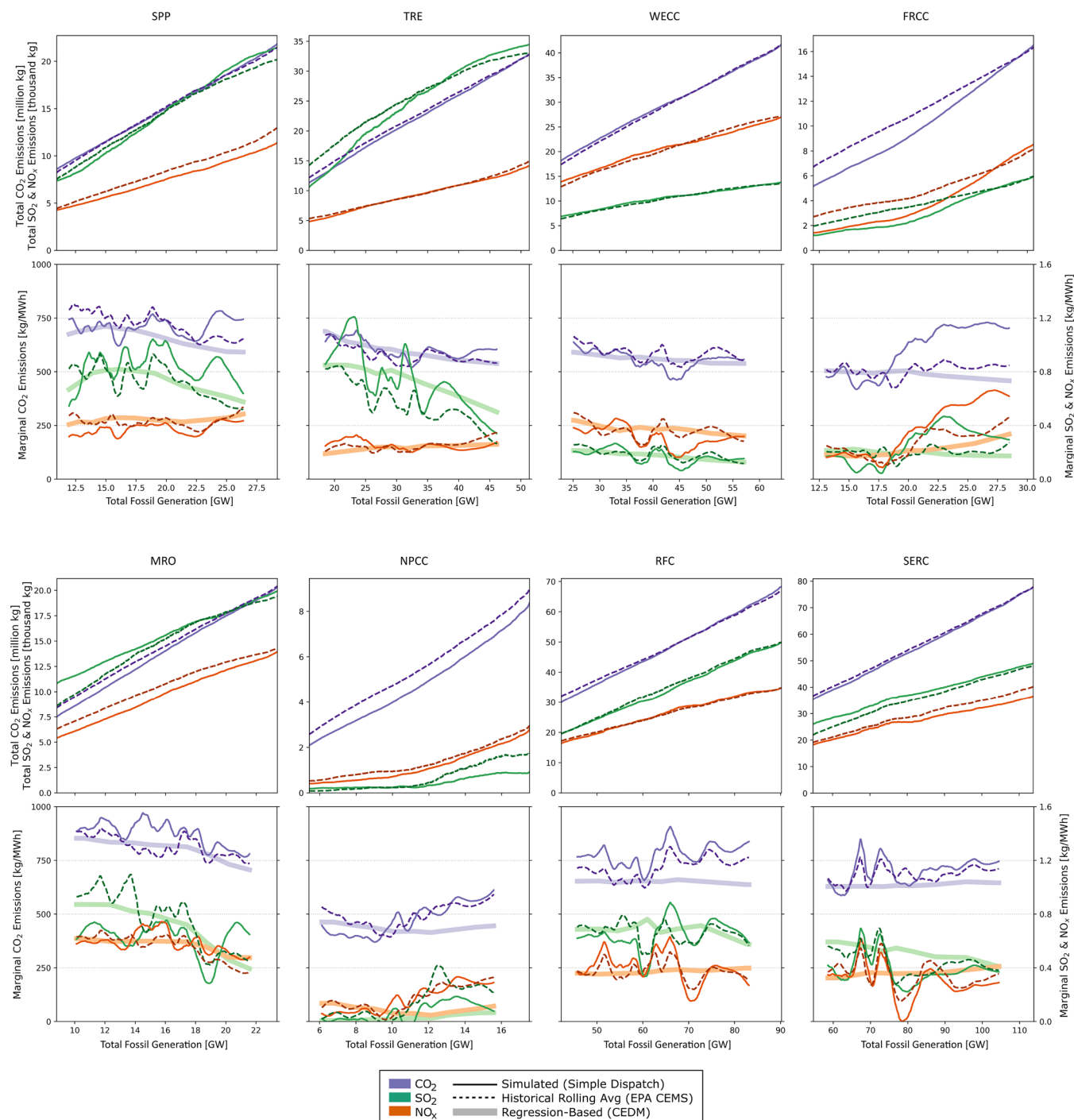


Figure 2. Total emissions and marginal emissions factors (MEFs) for all eight NERC regions in 2017. MEFs plots include regression-based MEFs from Azevedo et al.¹⁹ The x-axes cover the 5th to 95th percentiles of the total fossil generation. Since we calculate MEFs via a rolling average, the MEF curves cover the 10th to 90th percentile of the total fossil generation.

RESULTS AND DISCUSSION

Figure 1A illustrates the merit order curve for the WECC (Western), TRE (Texas), and RFC (East) regions during the first week of August 2017. The merit order curve will vary throughout the year with changes in heat rate, fuel price, power plant capacity, and unit maintenance.

Each of the eight NERC regions' power plant fleets has a distinct mixture of fuels and prime movers. In the WECC region, for example, we find bituminous and sub-bituminous coal plants provide baseload generation and natural gas tends to serve peak

demand. The TRE fleet shows a mixture of sub-bituminous, lignite, and natural gas combine cycle generators throughout the merit order. And RFC contains large amounts of coal generation, including refined coal, along with inexpensive baseload natural gas combined cycle generators. The Simple Dispatch model accommodates these varying power plant fleet mixtures.

Figure 1B shows the CO₂ emissions intensity of each power plant in Figure 1A. We find a wide variation in emissions intensity in plants that use the same fuel, even within the same

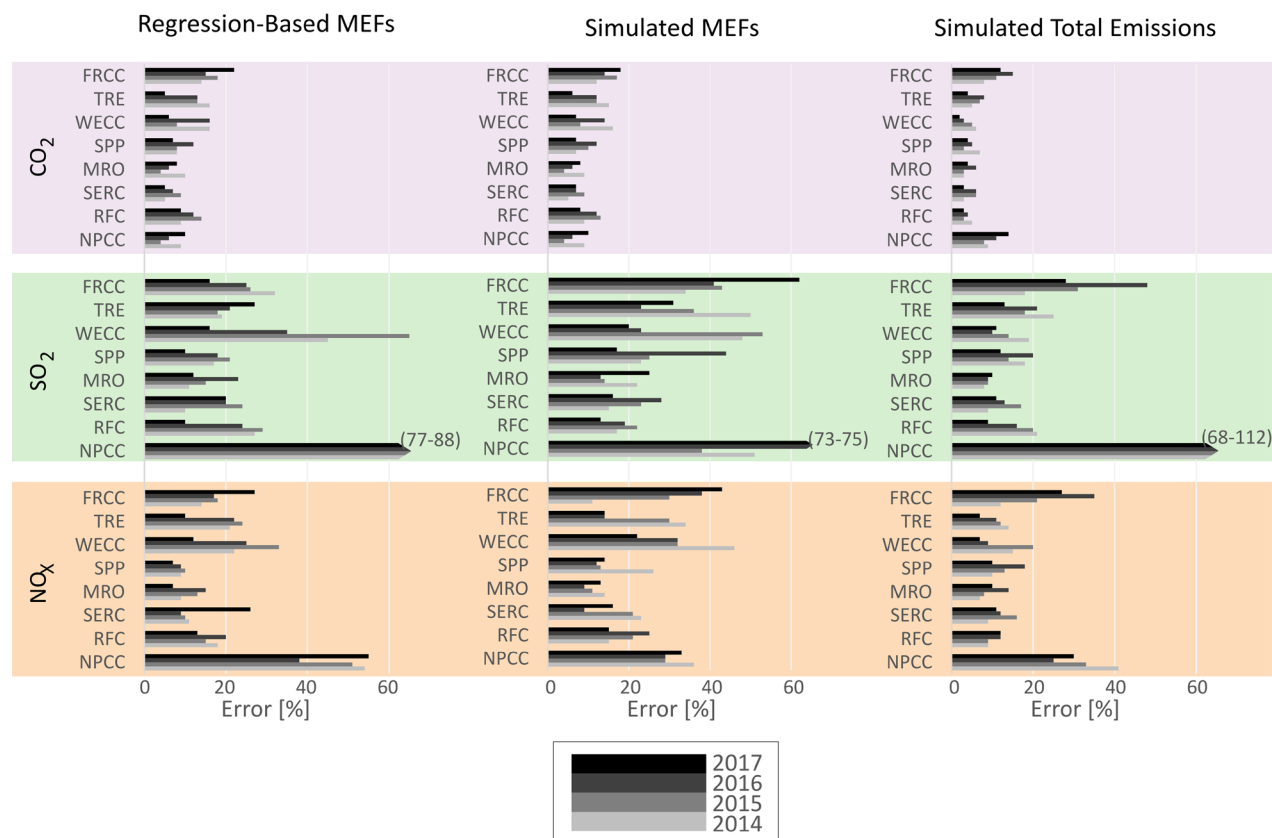


Figure 3. Relative error—see eq 3—for the regression-based MEFs,¹⁹ Simple Dispatch simulated MEFs, and Simple Dispatch simulated total emissions.

region. These emissions intensity merit order charts give some intuition about the relationship between total generation and simulated MEFs. In this particular week of the merit order, for example, we expect RFC's CO₂ MEFs to be greater with increasing total fossil generation because the upper portion of the merit order contains more high-emitting generators.

In Figure 2, we show the total and marginal emissions of CO₂, SO₂, and NO_x calculated from the Simple Dispatch simulation and from the 2017 historical data. We find good agreement between the simulated and historical data for all regions, especially for CO₂ and NO_x.

Figure 2 also includes the regression-based MEFs from Azevedo et al.¹⁹ The historical MEFs—the derivative of the historical curves of total emissions versus total fossil generation—tend to be more dynamic than the regression-based MEFs. The Simple Dispatch model captures these dynamics better than the regression-based model—especially in WECC, RFC, and SERC, for example. These quickly fluctuating MEFs arise when groups of low- or high-emitting power plants of similar production cost cluster in the same portion of the merit order.

The regression-based MEFs, however, are generally more accurate than the simulated values. Figure 3 shows the relative error—see eq 3—for each pollutant and region with the regression-based MEFs in the left panel and simulated MEFs in the middle. With few exceptions, the relative error between the simulated MEFs and the historical data exceeds the error for the regression-based MEFs. However, the error between the two models is comparable. The right panel of Figure 3 shows the relative error between Simple Dispatch and historical total

emissions—around 5% for CO₂, 10% for NO_x, and 15% for SO₂ in most regions.

For benchmarking, we also compare the 2014 TRE results against the output of a sophisticated optimization-based dispatch model implemented in the Plexos energy modeling software.¹⁵ The Plexos model simulates MEFs with less error than the Simple Dispatch method and more error than the regression-based method. The Plexos model simulates total emissions with greater error than the Simple Dispatch method. See Supporting Information for more detail. Using a sophisticated optimization-based model will likely lead to more accurate simulated MEFs, but the relative error of the MEFs produced by the Simple Dispatch model remain in an acceptable range. The Simple Dispatch model also solves in 30 min on a personal laptop computer versus the 20 h required to solve the Plexos model on a high-performance desktop machine.

SO₂ emissions are the most difficult of the pollutants to simulate accurately. This discrepancy stems from the great variety in SO₂ emissions rates among power plants and across seasons. SO₂ emissions are also driven primarily by coal; any minor simulation errors with the coal fleet dispatch can produce large errors in simulated SO₂ emissions.

FRCC and NPCC are the most difficult regions to simulate accurately. Both regions have relatively small coal fleets; any minor errors in each coal plant's position in the merit order could yield large errors in simulated emissions. Our zero-transmission-congestion assumption may also contribute to the error if these regions experience chronic transmission constraints that affect the power plant dispatch.

In conclusion, we developed an open-source, reduced-order model that can accurately analyze broad trends affecting the

economic dispatch of regional power plant fleets—for example, calculating marginal emissions factors. The method's simplicity enables it to quickly analyze scenarios for multiple regions and years at an hourly resolution. Its simulation-based framework enables it to analyze future scenarios that are absent from the historical data. Historical regression-based MEFs, however, are generally more accurate than the MEFs simulated in our model. Though our simulated MEFs present a more nuanced picture of how clusters of low- or high-emitting power plants of similar production cost create large swings in MEFs as we ascend the merit order. The marginal emissions factors estimated in this paper are available for download at <https://cedm.shinyapps.io/MarginalFactors/>. Policymakers could use these dynamic MEFs to target demand-reduction strategies at high-emissions portions of the power plant merit order.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.est.9b02500.

URL location for downloading the open-source model scripted in the Python programming language; map of the NERC region boundaries; detailed information on variable operation and maintenance (VOM) costs; description of how the minimum downtime constraint is implemented in the model; discussion of NERC inter-regional trade and potential influence on the Simple Dispatch solution; comparison of error for the Plexos, Simple Dispatch, and regression-based simulation methods; validation of the model's simulated prices (PDF)

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Notes

The authors declare no competing financial interest.

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